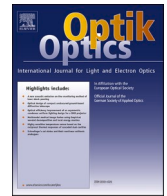




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A critical review and future prospects of Dy³⁺-doped glasses for white light emission applications

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ARTICLE INFO

Keywords:

Dy³⁺:glasses
Luminescence
Judd-Ofelt theory
Y/B ratios
Color coordinates

ABSTRACT

Rare-earth (RE)-doped glassy systems gained much interest because of its wide variety of applications in numerous potential fields that include optical fibers, solid state lasers, optical amplifiers, optical thermometry sensors, waveguides, infrared detectors, spectral converters, display devices, and so on. Within RE ions, the trivalent dysprosium (Dy³⁺) ion is the most unique and characteristic ion to examine emission properties for white light and visible display emitting devices. Because, Dy³⁺ ion emits two intense emissions in the blue (~480 nm) and yellow (~575 nm) regions that are assigned to the ⁴F_{9/2}→⁶H_{15/2} and ⁴F_{9/2}→⁶H_{13/2} transition energies, respectively. Under the excitation of blue laser, Dy³⁺ ions exhibit attractive yellowish white fluorescence emission with higher quantum efficiency. In view of the importance of identifying efficient luminescent emitting materials for the development of optical devices, the present work is to investigate white light emission in new glass composition of B₂O₃ +TeO₂ +SrCO₃ +ZnO+ZnF₂ +Dy₂O₃ glass and in turn to compare and discuss these results with reported Dy³⁺:glasses. The tabulated reported data bank for 68 Dy³⁺:glasses at one place should prove valuable information for future work on rationalization of theoretical and practical point of view to enrich white light emission quantification, simulation and systemization.

1. Introduction

Now-a-days the rare earth (RE)-embedded glasses are excellent visible emitting devices due to the emission of intense and sharp luminescence in infrared (IR), visible (VIS) and ultraviolet (UV) spectral regions because of the shielding of 4f electrons by the filled orbitals of 5s and 5p [1–10]. Such glasses give rise to potential scientific and technological applications through efficient 4f-5d and 4f-4f electronic transitions of RE ions [1,2,5–7,9]. It is well known that the spectroscopic characteristics of the RE³⁺-glasses robustly depends on the geometry and nature of the host matrix and these properties can be tailored by the selection of appropriate host glass network formers and modifiers. In recent past, the quest for investigation of appropriate materials for luminescent devices has yielded

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some valuable and useful scientific reports but still there is a chance to do rigorous research in the field of RE-doped glasses.

Glasses are one of the attractive and extensively used materials in diverse fields. The attractive feature of glasses include broad range of composition, easy fabrication in any shapes, homogeneity and size. RE ions doped glasses are found to be excellent candidate for the development of optical sensors and lasers due to their efficient emission properties in visible, infrared, and near infrared spectral regions.

Among trivalent rare earth (RE^{3+}) elements, Dy^{3+} ion possess its own characteristic spectral features that include emission of blue and yellow (red+green) light at $\sim 470\text{--}490\text{ nm}$ ($^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$) and $\sim 555\text{--}595\text{ nm}$ ($^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$), respectively, which combine together yields white light. However, perfect white light depends on the emission intensity ratio between blue (B) and yellow (Y) which in turn depends on the host matrix nature and geometry. Also, infrared emission at $1.3\text{ }\mu\text{m}$ ($^6\text{F}_{11/2} \rightarrow ^6\text{H}_{15/2}$) of Dy^{3+} is very much useful to enhance the efficiency of the dysprosium doped optical fiber amplifiers. Therefore, visible emission as well as infrared emission can be tuned and tailored by changing and selecting proper host. Among different hosts of glasses, glass-ceramics and crystalline environments, glassy media is the best choice. In this direction, variety of oxide glass hosts such as borates [11–40], silicates [41–45], tellurites [46–55], germanates [2,56] and phosphate [57–64] based glasses doped with Dy^{3+} have been studied to optimize and enhance the white light and infrared emission.

Among oxide glasses, borate glasses may be preferred due to their homogeneity and transparency, excellent thermal stability and chemical durability besides high Dy^{3+} ion doping [11–40]. However, high phonon energy ($\sim 1100\text{ cm}^{-1}$) of borate glasses greatly reduces the radiative properties through increased multiphonon relaxation [11–40]. In order to reduce phonon energy, either adding one more glass former or glass network modifiers have been used in host glass composition. In this direction TeO_2 (another glass former) and network modifiers such as fluorides to reduce the phonon energy as well as zinc/strontium oxides have been added to improve the mechanical strength and ability to form the glass [18,19,21,22].

At present there is a huge demand of good optical quality systems for solid state visible lighting, w-LEDs, waveguides, high density optical memories based on frequency-domain, lasers, and optical amplifiers. In this connection, widely being focused on the study of optical properties of RE-doped silicate, tellurite, germanate, and phosphate glasses. The main objectives of this paper is to study the emission properties of new glass composition ($\text{B}_2\text{O}_3 + \text{TeO}_2 + \text{SrCO}_3 + \text{ZnO} + \text{ZnF}_2 + \text{Dy}_2\text{O}_3$). The characteristic emission properties obtained in the present work have been discussed along with the reported 68 Dy^{3+} :glasses [11–64].

2. Experimental

A series of required dimension of glasses with the batch of $(40-x)\text{B}_2\text{O}_3 + 25\text{TeO}_2 + 15\text{SrCO}_3 + 10\text{ZnO} + 10\text{ZnF}_2 + x\text{Dy}_2\text{O}_3$ (where $x = 0.1, 0.25, 0.5, 1$ and $2\text{ mol\%}$) have been prepared following well known melt and rapid quenching method and the resulted glass pieces were labeled as $\text{BTSrZofDy}_{0.1}$, $\text{BTSrZofDy}_{0.25}$, $\text{BTSrZofDy}_{0.5}$, $\text{BTSrZofDy}_{1.0}$ and $\text{BTSrZofDy}_{2.0}$, respectively, as proposed by Chandrappa et al. [65] and Ramprasad et al. [66]. The required precursor raw chemicals (99.99%) were purchased through M/s.Sigma Aldrich Ltd. Each batch was melted at $920\text{ }^\circ\text{C}$ for 1.5 hrs to get bubble free with clear molten melt and then quenched very rapidly in a predetermined mould and quickly transferred to another furnace and annealed for 9 hrs at $300\text{ }^\circ\text{C}$. Later as-quenched glasses are polished on both sides to keep uniformity.

The density and refractive index values of the glasses are measured by employing Archimedes method (Xylene as immersion liquid) and Abbe's refractometer (sodium vapour lamp source at wavelength 589.3 nm), respectively. The absorption studies were carried out using JASCO UV-Vis-NIR V-670 spectrophotometer. The photoluminescence characteristics of the studied glasses are measured by using Perkin Elmer spectrometer and the lifetime measurements are done by using F980 fluorescence spectrometer.

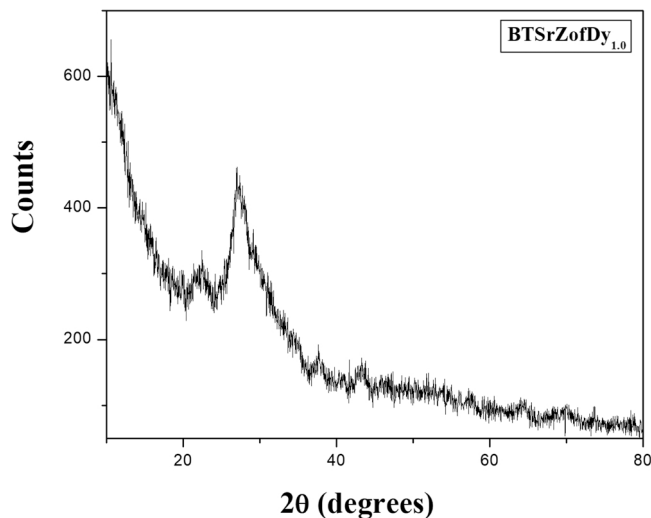


Fig. 1. XRD pattern of the $\text{BTSrZofDy}_{1.0}$ glass.

3. Results and discussion

The XRD pattern for the BTsZofDy_{1.0} glass is shown in Fig. 1 which clearly confirms amorphous nature without any crystal phase formation with extensive dispersal peak and the nonexistence of characteristic narrow diffraction peaks. Also, for undoped and doped with all concentrations of Dy³⁺ glasses exhibit similar nature of XRD pattern except minor changes in intensities. Various physical properties of the Dy³⁺-doped BTsZofDy glasses which are essential to find out the required optical and luminescence properties are calculated and are tabulated in Table 1. Fig. 2 shows the observed absorption spectrum (400–1800 nm) along with 9 transition labels for BTsZofDy_{2.0}. The spectrum shows nine bands and the assignments of the bands are also shown in Fig. 2 and collected in Table 2. The Dy³⁺ ion of [Xe]4f⁹5s²5p⁶ configuration gives rise many excited electronic states originating from ground ⁶H_{15/2} state. Only minor variation in peak positions as well as intensities of the absorption energies are noticed compared to the reported Dy³⁺ glasses [15,23,25]. The bonding parameter (δ) by using Nephelauxetic ratio (β) are estimated from the observed energies in a given host and free-ion (aquo-ion) energy levels [67] and these values are tabulated in Table 2. The bonding parameter ' δ ' [68,69] is defined as

$$\delta = \left[\frac{1 - \beta}{\beta} \right] \times 100 \text{ and } \beta = \nu_c / \nu_a \quad (1)$$

where $\bar{\beta} = \sum \frac{\beta}{N}$, ' β ' refers the Nephelauxetic ratio. Here ν_c and ν_a refers the energies of the corresponding transition in the complex (BTsZofDy) and aquo-ions, respectively. ' N ' refers the number of levels used to compute ' β ' values. The ' δ ' may be positive or negative indicating covalent or ionic bonding. The Dy³⁺ ions experiences ionic nature in the present host glass due to negative value of ' δ '.

The oscillator strengths of experimental (f_{exp}) and calculated (f_{cal}) are evaluated and are shown in Table 1 along with JO parameters [70,71] as carried out following usual procedure [1–3], using reduced matrix elements reported in ref. [72]. In the present Dy³⁺-doped tellurofluoroborate glasses, the trend in JO parameters has been observed as $\Omega_2 > \Omega_6 > \Omega_4$. The larger Ω_2 indicates the strong interactions between Dy³⁺ ion and neighboring atoms with stronger covalency.

Branching ratios (β_R), total radiative transition probabilities (A_T) and radiative lifetimes (τ_{rad}) have been predicted with the help of JO parameters for the excited fluorescent level of ⁴F_{9/2} level of Dy³⁺ ion and are presented in Table 3 [28,40,42]. The ⁴F_{9/2} level possess purely radiative relaxation as this level has sufficient energy gap to its immediate lower energy level, ⁶F_{3/2}.

The excitation spectrum of Dy³⁺:BTsZofDy_{0.5} glass is shown in Fig. 3 which is more or less similar to other reported Dy³⁺ glasses [45,46,56,57]. As seen from the figure, the excited energy levels (emission at 575 nm) peaked at 325 nm (⁶H_{15/2} → ⁶P_{5/2}), 350 nm (⁶H_{15/2} → ⁶P_{7/2}), 375 nm (⁶H_{15/2} → ⁶P_{3/2}), 387 nm (⁶H_{15/2} → ⁴I_{13/2}), 425 nm (⁶H_{15/2} → ⁴G_{11/2}), 451 nm (⁶H_{15/2} → ⁴I_{15/2}) and 475 nm (⁶H_{15/2} → ⁴F_{9/2}) are identified. It is also observed that the spectral profile (shape and peak position) found to be independent on Dy³⁺ ion concentration. The characteristic peaks near UV region (315–405 nm) can be highly useful for excitation and in turn to get efficient emission of Dy³⁺:BTsZofDy. Emission spectra from ⁴F_{9/2} to ⁶H_{15/2,13/2,11/2} have been measured by exciting ⁶H_{15/2} → ⁴I_{13/2} (387 nm) which are responsible for characteristic white light emission in Dy³⁺:BTsZofDy [40,41,46,47]. The luminescence spectra (excitation at 387 nm) for Dy³⁺ (0.1–2.0 mol%) embedded tellurofluoroborates are shown in Fig. 4. All emission bands in Fig. 3 are attributed to ⁴f⁹–⁴f⁹ (Dy³⁺) transitions. Under 387 nm excitation, emissions of blue (located at 485 nm) and yellow (centered at 576 nm) besides weak red (peaked at 664 nm) were noticed and are labeled as ⁴F_{9/2} → ⁶H_{15/2}, ⁶H_{13/2} and ⁶H_{11/2}, respectively, of Dy³⁺ ions. It is also noticed that the emission levels of Dy³⁺ ions are broadened due to average splitting of Stark levels between ⁴F_{9/2} and ⁶H_J multiplets.

Table 1

Experimental and calculated oscillator strengths ($\times 10^{-6}$), rms deviation (σ), Judd-Ofelt ($\times 10^{-20} \text{ cm}^2$) parameters, density ($d, \text{ g/cm}^3$) and refractive index (n) of the Dy³⁺ ions doped zinc boro-tellurite glasses.

Transition from ⁶ H _{15/2}	BTsZofDy _{0.1}		BTsZofDy _{0.25}		BTsZofDy _{0.5}		BTsZofDy _{1.0}		BTsZofDy _{2.0}	
	f_{exp}	f_{cal}	f_{exp}	f_{cal}	f_{exp}	f_{cal}	f_{exp}	f_{cal}	f_{exp}	f_{cal}
⁶ F _{3/2}	0.42	0.24	0.19	0.26	0.12	0.21	0.16	0.21	0.11	0.18
⁶ F _{5/2}	1.26	1.26	1.23	1.38	1.53	1.09	1.32	1.13	1.55	0.98
⁶ F _{7/2}	2.15	2.44	2.42	2.67	2.46	2.29	2.48	2.28	1.98	2.11
⁶ F _{9/2}	2.10	2.03	2.27	2.22	2.33	2.38	2.08	2.13	2.34	2.32
⁶ F _{11/2}	6.41	6.43	6.21	6.23	6.90	6.87	5.16	5.14	5.20	5.19
⁶ H _{11/2}	1.64	1.50	1.75	1.58	1.18	1.38	1.13	1.29	1.07	1.16
N	6		6		6		6		6	
σ	±0.155		±0.140		±0.213		±0.132		±0.246	
Ω_2	6.06		5.66		6.25		4.26		4.11	
Ω_4	0.19		0.18		1.25		0.71		1.47	
Ω_6	2.78		3.05		2.39		2.48		2.14	
d	3.554		3.119		3.686		3.826		4.318	
n	1.688		1.682		1.694		1.699		1.701	

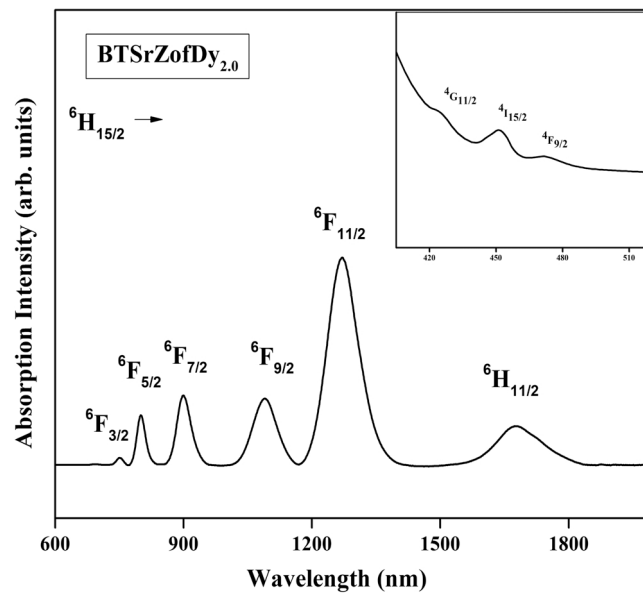


Fig. 2. Absorption spectrum of the BTSrZofDy_{2.0} glass in the Vis-NIR region.

Table 2

Observed band positions (cm⁻¹), bonding parameters ($\bar{\beta}$ and δ) of the Dy³⁺ ions doped zinc boro-tellurite glasses.

Transition from ⁶ H _{15/2}	BTSrZofDy _{0.1}	BTSrZofDy _{0.25}	BTSrZofDy _{0.5}	BTSrZofDy _{1.0}	BTSrZofDy _{2.0}	Aquo-ion
⁶ F _{3/2}	13316	13333	13316	13316	13333	13250
⁶ F _{5/2}	12500	12500	12547	12500	12500	12400
⁶ F _{7/2}	11148	11148	11148	11148	11148	11000
⁶ F _{9/2}	9183	9208	9217	9183	9183	9100
⁶ F _{11/2}	7874	7855	7855	7868	7874	7700
⁶ H _{11/2}	6002	5999	5999	5988	5988	5850
$\bar{\beta}$	1.0140	1.0142	1.0149	1.0135	1.0139	–
δ	– 1.3845	– 1.4024	– 1.4572	– 1.3316	– 1.3664	–

Table 3

Peak wavelength (λ_p , nm), effective linewidth ($\Delta\lambda_{\text{eff}}$, nm), radiative transition probability (A , s⁻¹), peak stimulated emission cross-section ($\sigma(\lambda_p)$, $\times 10^{-21}$ cm²), experimental (β_{exp}) and calculated (β_{cal}) branching ratios experimental (τ_{exp} , ms) and calculated (τ_{rad} , ms) lifetimes and quantum efficiency (η , %) of the Dy³⁺ ions doped zinc boro-tellurite glasses.

Emission transitions	Parameters	BTSrZof Dy _{0.1}	BTSrZof Dy _{0.25}	BTSrZof Dy _{0.5}	BTSrZofDy _{1.0}	BTSrZofDy _{2.0}
⁴ F _{9/2} → ⁶ H _{15/2}	λ_p	481	482	482	482	483
	$\Delta\lambda_{\text{eff}}$	0.747	0.410	0.457	0.576	0.648
	A	257	278	240	243	223
	$\sigma(\lambda_p)$	8.58	17.16	13.10	10.44	8.59
	β_R (Exp)	0.372	0.340	0.347	0.415	0.360
	β_R (Cal)	0.196	0.213	0.180	0.217	0.206
	λ_p	574	574	575	574	575
⁴ F _{9/2} → ⁶ H _{13/2}	$\Delta\lambda_{\text{eff}}$	0.470	0.387	0.424	0.417	0.422
	A	806	787	825	649	626
	$\sigma(\lambda_p)$	86.70	103.50	127.15	77.54	74.35
	β_R (Exp)	0.585	0.590	0.611	0.534	0.594
	β_R (Cal)	0.615	0.602	0.620	0.581	0.579
	λ_p	664	663	664	663	662
	$\Delta\lambda_{\text{eff}}$	0.359	0.569	0.502	0.686	0.406
⁴ F _{9/2} → ⁶ H _{11/2}	A	91	87	95	73	72
	$\sigma(\lambda_p)$	23.06	13.85	17.04	9.49	15.59
	β_R (Exp)	0.043	0.071	0.042	0.047	0.046
	β_R (Cal)	0.070	0.066	0.072	0.066	0.067
	τ_{rad} (ms)	0.763	0.765	0.730	0.895	0.824
	τ_{exp} (ms)	0.524	0.398	0.336	0.237	0.157
	η (%)	60.68	52.03	56.16	65.59	41.87

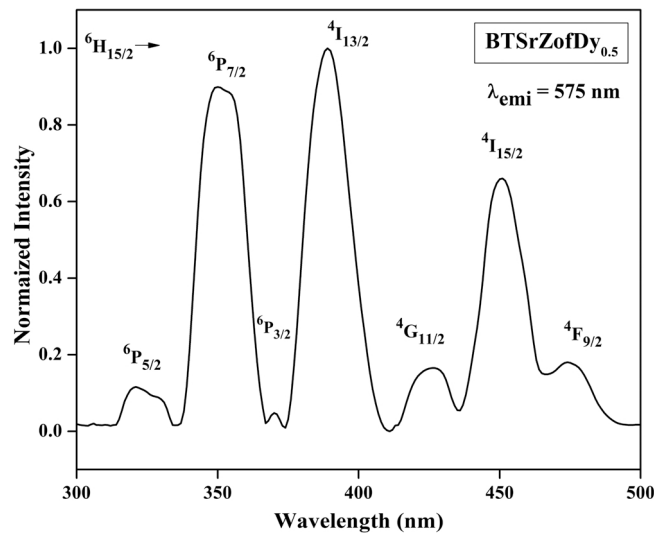


Fig. 3. Excitation spectrum of the Dy^{3+} ions doped zinc boro-tellurite glasses.

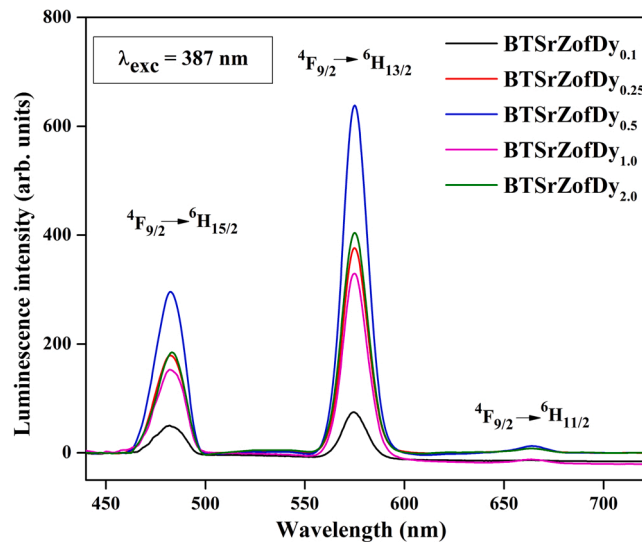


Fig. 4. Emission spectra of the Dy^{3+} ions doped zinc boro-tellurite glasses.

The change in peak intensities of ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{15/2}$, ${}^6\text{H}_{13/2}$ and ${}^6\text{H}_{11/2}$ transitions increases with increase of Dy^{3+} ion concentration up to 0.5 mol% and then decreased upto 2.0 mol%, as observed in Fig. 4. The emission intensities are quenched due to the addition of Dy^{3+} ion concentration beyond 0.5 mol%. The decrease in intensity beyond 0.5 mol% was ascribed either resonant energy transfer (RET) from the excited ${}^4\text{F}_{9/2}$ to ${}^6\text{H}_{15/2}$ ground levels or cross-relaxation (CR) phenomenon among acceptor and donor ions.

Several spectral parameters (peak position (λ_p), band width ($\Delta\lambda_{\text{eff}}$), lifetime (τ_{exp}), total radiative transition probability (A_T) and stimulated emission cross-section ($\sigma(\lambda_p)$) for Dy^{3+} :BTSrZofDy glasses were estimated by using luminescence data and are tabulated in Table 3 [27,42,46,51]. The hypersensitive (${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{13/2}$) yellow (Y) luminescence intensity mainly varies with the host, in comparison to the less dependent blue (B) ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{15/2}$ (Dy^{3+}) transition and therefore Y/B ratio can be modulated to get optimum white light generation. The experimental τ_{exp} and $\sigma(\lambda_p)$ are the most essential parameters required to access for solid-state laser materials. The $\sigma(\lambda_p)$ are considerably larger for ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{13/2}$ transition of Dy^{3+} ion than for remaining transitions. The assessed Y/B quantities of BTSrZofDy are shown in Table 4. As noticed from Table 4, Y/B ratio varied within white light region from 1.572 to 1.761 while changing the Dy^{3+} concentrations, confirming the stability in white light generation irrespective of Dy^{3+} content.

The chromaticity coordinates evaluated for BTSrZofDy glasses are tabulated in Table 4 and also shown in Fig. 5. From the Commission Internationale de L'Eclairage plot, it is noticed that the BTSrZofDy glasses emit white light under the excitation of 387 nm and therefore found to be used for the development of white LEDs.

The quality of white light was estimated with the help of McCamy's approximation as follows [73],

Table 4

Y/B ratio, Chromaticity color coordinates (x, y), Correlated color temperature (CCT) and Color purity of the Dy³⁺ ions doped zinc boro-tellurite glasses.

Glass code	Y/B	Chromaticity coordinates		CCT	Color purity
		x	y		
BTSrZofDy _{0.1}	1.572	0.295	0.325	7611	0.104
BTSrZofDy _{0.25}	1.739	0.339	0.374	5262	0.131
BTSrZofDy _{0.5}	1.761	0.324	0.353	5841	0.069
BTSrZofDy _{1.0}	1.285	0.353	0.381	4829	0.161
BTSrZofDy _{2.0}	1.650	0.304	0.336	6911	0.077

$$\text{CCT} = -449n^3 + 3525n^2 - 6823n + 5520.33 \quad (2)$$

Where CCT refers correlated color temperature, 'n' concerns inverse slope $(x-x_e)/(y-y_e)$ and $(x_e = 0.332, y_e = 0.186)$ indicates the epicenter of the iso temperature lines. Determined CCT values of BTSrZofDy glasses are shown in Table 4 and ranges between 4829 and 7611 K confirming the emission of cool (CCT > 4000 K) white light. It is also noted that even for different excitation, CCT estimated to be more than 4000 K which also confirms that BTSrZofDy glasses can emit cool white light and therefore be useful for cool wLEDs applications.

The decay rates for the $^4F_{9/2}$ level of Dy³⁺:BTSrZofDy glasses have been measured, exciting at 387 nm, and presented in Fig. 6. As can be observed from Fig. 6, the decay rates are in mono-exponential nature for lower and then it turns into bi-exponential for higher Dy³⁺ ion concentration. The values of τ_{exp} ($^4F_{9/2}$ level of Dy³⁺) are determined as 0.524, 0.398, 0.336, 0.237 and 0.157 ms for BTSrZofDy_{0.1}, BTSrZofDy_{0.25}, BTSrZofDy_{0.5}, BTSrZofDy_{1.0} and BTSrZofDy_{2.0} glasses, respectively, reveals that the τ_{exp} shortened with increase in Dy³⁺ ions.

The decay curves reveal mono-exponential trend for lower Dy³⁺ (0.1 and 0.25 mol%) content, on the other hand, for higher Dy³⁺ (≥ 0.5 mol%) content, the decay curve systematically change towards a bi-exponential trend accompanied by shortening of lifetime. This decrease in lifetime is mainly due to decrease in the separation between ion and ion which leads the transfer from excited (donor) ion to acceptor (de-excited ion), involving pair of excited and ground states. This type of energy transfer mechanism leads bi-exponential trend of the decay times. Overall, at low RE ions content, the luminescence decay curves agree to fit a mono-exponential where the interaction due to optically active ions is not at all significant. The energy transfer mechanism can be identified by analysing the well-known theoretical Inokuti-Hirayama (IH) model [74], the bi-exponential trend due to multipolar interaction, the decay of emission intensity $I(t)$ that can be expressed with time dependence is as follows:

$$I(t) = I_0 \exp \left[-\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{\frac{2}{3}} \right] \quad (3)$$

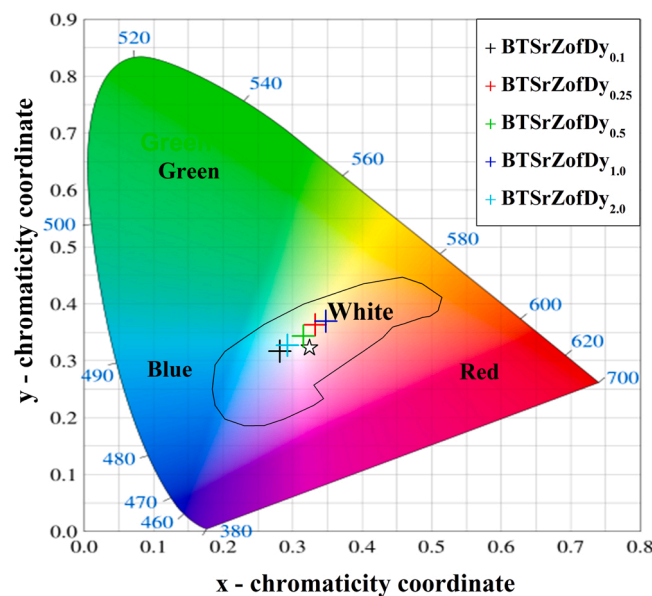


Fig. 5. CIE 1931 diagram of the Dy³⁺ ions doped zinc boro-tellurite glasses.

where τ_0 relates the intrinsic decay rate of the ${}^4F_{9/2}$ (Dy^{3+}) state estimated for the lowest content of Dy^{3+} whereas 'Q' refers energy transfer quantity and 'S' refers the multipolar interaction involved, depending on S = 6, 8 or 10 corresponds to dipole-dipole, dipole-quadrupole or quadrupole-quadrupole, correspondingly. In the present study, the predominant interaction between Dy^{3+} and Dy^{3+} has been well fitted for S = 6 and 'Q' is found to be 0.53, 1.34 and 3.27 for 0.5, 1.0 and 2.0 mol% of Dy^{3+} :BTsZofDy glasses, respectively. Hence, it is concluded that the energy transfer process ($Dy^{3+} \rightarrow Dy^{3+}$) takes place through interaction based on electric dipole-dipole. Based on energy transfer quantity (Q), acceptor coupling constant (C_{DA}) and critical transfer distance (R_0) are estimated. The C_{DA} values are found to be 9.02×10^{-38} , 49.13×10^{-38} and $425 \times 10^{-38} \text{ cm}^2 \text{ s}^{-1}$ and R_0 values are 5.3 Å, 7 Å and 10.15 Å for 0.5, 1.0 and 2.0 mol%, respectively. It is clear that the magnitude of 'Q' is increasing and critical distance between the donor (excited Dy^{3+}) and acceptor (non-excited Dy^{3+}) also increases with increase in Dy^{3+} ion concentration.

The non-radiative rates of ${}^4F_{9/2}$ level can be determined by the following relation [42,46],

$$\frac{1}{\tau_{\text{exp}}} = \frac{1}{\tau_{\text{rad}}} + W_{\text{NR}} \quad (4)$$

where W_{NR} refers probability due to non-radiative relaxation which is the sum of energy transfer probability (W_{ET}) and multiphonon relaxation rate (W_{MP}) where later depends on the phonon energy of the host whereas the former depends on active (doping) ion concentration. For low concentration, W_{ET} can be ignored. On the other hand, W_{MP} depends on the energy gap between the excited fluorescent state and the next adjacent lower state besides the highest vibration energy of the host. In the case of Dy^{3+} :glasses, W_{MP} processes are neglected due to the large energy gap between the emitting levels to the next lower level. As such, from Eq. (4), W_{NR} can be determined by using the τ_{exp} and τ_{rad} , which gives nearly 514 s^{-1} (BTsZofDy_{1.0}). For higher concentration of Dy^{3+} :BTsZofDy, shortening of lifetimes leads bi-exponential trend and are explained to the increase in energy transfer with Dy^{3+} ions. This observation reflects the decrease in the Dy^{3+} - Dy^{3+} distances at higher Dy^{3+} ion concentration in BTsZofDy.

Each glass composition has its own influence on luminescence properties and therefore each characteristic property varies are varying with the variation of glass host. Due to importance and the significance of each parameter as well as dependence of the host these values are compared for 68 Dy^{3+} -doped systems to explore possible relation/logic between Dy^{3+} ion-ligand field and white light emission. The luminescence properties of τ_{rad} , τ_{exp} , quantum efficiency (η), W_{NR} of ${}^4F_{9/2}$ level; peak emission wavelengths (λ_p) line-widths ($\Delta\lambda_{\text{eff}}$), emission cross-sections ($\sigma(\lambda_p)$), gain bandwidth ($\sigma(\lambda_p) \times \Delta\lambda_{\text{eff}}$, $\times 10^{-28} \text{ cm}^2$), optical gain ($\sigma(\lambda_p) \times \tau_R$, $10^{-24} \text{ cm}^2 \text{ s}$) and figure of merit ($\sigma(\lambda_p) \times \tau_{\text{exp}}$, $10^{-24} \text{ cm}^2 \text{ s}$) for ${}^4F_{9/2} \rightarrow {}^6H_{15/2}$ and ${}^6H_{13/2}$ transitions of 1.0 mol% Dy_2O_3 doped glass of the present investigation are compared with 65 reported glasses [11–64] doped with 1.0 mol/% Dy_2O_3 and are shown in Table 5. Readers are advised to refer our earlier paper for the details of glass composition and glass labels [65] used in Table 5. As seen from Table 5, maximum quantum efficiency (more than 90%) for TWTiDy_{1.0} [47], BABfPfDy_{1.0} [21], TPofZNDy_{1.0} [50], TWKfLfDy_{1.0} [51], TWKfDy_{1.0} [51], BLMBfNfDy_{1.0} [26], BLCBfNfDy_{1.0} [26], BTsABCdDy_{1.0} [30], BTNBfCdDy_{1.0} [31] and PBiLNDy_{1.0} [62] where as less efficiency (less than 30%) for BTsZofDy_{1.0} [P.W.], TZNDy_{1.0} [46], BTAKDy_{1.0} [17] and SBNABaZDy_{1.0} [41] has been noticed.

Fig. 7 shows the τ_{exp} and W_{NR} for ${}^4F_{9/2}$ level versus different 1.0 mol% of Dy^{3+} :glasses. In the x-axis, in the first instant, the compositions are arranged and fixed with increasing order of W_{NR} , then the values of τ_{exp} are taken for respective compositions. In x-axis, the serial numbers 1–15 are taken as 1. BABfPfDy_{1.0} [21], 2. BLBiGdDy_{1.0} [28], 3. BTBiZbDy_{1.0} [20], 4. BTsZofDy_{1.0} [P.W.], 5. BTCAMTiDy_{1.0} [19], 6. PZNofAfDy_{1.0} [60], 7. BSLZfDy_{1.0} [12], 8. BPKAZfDy_{1.0} [14], 9. BPbTPBaZDy_{1.0} [27], 10. TBZofBfCdDy_{1.0} [49], 11. BZABfCdDy_{1.0} [38], 12. BPbAZDy_{1.0} [22], 13. SBNABaZDy_{1.0} [41], 14. TWLfNfDy_{1.0} [51] and 15. TWLfDy_{1.0} [51]. As seen from Fig. 7, there is no common trend of variation between W_{NR} and τ_{exp} with change in glass composition.

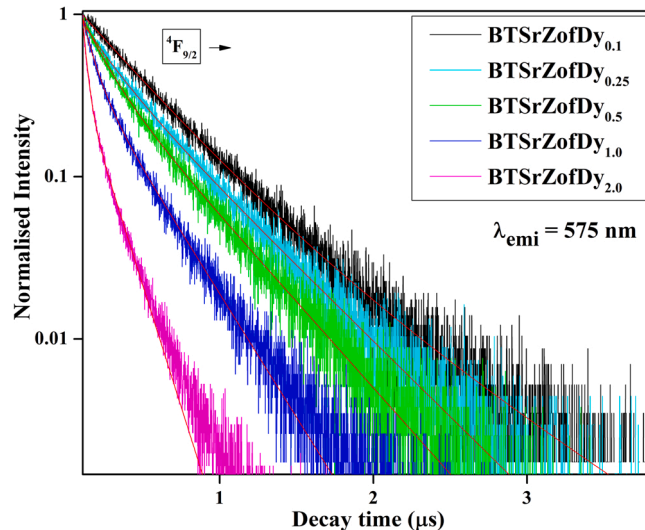


Fig. 6. Decay profile of the Dy^{3+} ions doped zinc boro-tellurite glasses corresponding to the ${}^4F_{9/2}$ energy level.

Table 5

Radiative (τ_{rad}) and experimental (τ_{exp}) lifetimes (ms), quantum efficiency (η), non-radiative relaxation (W_{NR}) for $^4\text{F}_{9/2}$ level, Peak wavelength (λ_p , nm), effective linewidth ($\Delta\lambda_{\text{eff}}$, nm), peak stimulated emission cross-section ($\sigma(\lambda_p)$, $\times 10^{-21}\text{cm}^2$), gain bandwidth ($\sigma(\lambda_p) \times \Delta\lambda_{\text{eff}}$, $\times 10^{-28}\text{cm}^3$), optical gain ($\sigma(\lambda_p) \times \tau_{\text{rad}}$, $10^{-24}\text{cm}^2\text{s}$) and figure of merit ($\sigma(\lambda_p) \times \tau_{\text{exp}}$, $10^{-24}\text{cm}^2\text{s}$) for $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$ and $^6\text{H}_{13/2}$ transitions of 1.0 mol/% Dy_2O_3 doped glasses.

	$^4\text{F}_{9/2}$				$^6\text{H}_{15/2}$						$^6\text{H}_{13/2}$						
	Glass Label	τ_{rad}	τ_{exp}	η	W_{NR}	λ_{p}	$\Delta\lambda_{\text{eff}}$	$\sigma(\lambda_{\text{p}})$	$\sigma(\lambda_{\text{p}}) \cdot \Delta\lambda_{\text{eff}}$	$\sigma(\lambda_{\text{p}}) \cdot \tau_{\text{rad}}$	$\sigma(\lambda_{\text{p}}) \cdot \tau_{\text{exp}}$	λ	$\Delta\lambda$	$\sigma(\lambda_{\text{p}})$	$\sigma(\lambda_{\text{p}}) \cdot \Delta\lambda_{\text{eff}}$	$\sigma(\lambda_{\text{p}}) \cdot \tau_{\text{rad}}$	$\sigma(\lambda_{\text{p}}) \cdot \tau_{\text{exp}}$
α	BTSrZofDy _{1.0} [PW]	895	237	26	3102.1	482	0.576	10.44	60.13	9.3438	2.4742	574	0.417	77.54	3.2334	6.9398	1.8376
	TZNDy _{1.0} [46]	483	145	30	4826.16	–	12.26	59.7	73.1922	2.88351	0.86565	–	10.83	45.2	48.9516	2.18316	0.6554
	TN10ZofDy _{1.0} [46]	267	166	62	2278.78	–	12.26	8.6	10.5436	0.22962	0.14276	–	10.91	81.3	88.6983	2.17071	1.34958
	TN20ZofDy _{1.0} [46]	326	163	50	3067.48	–	12.79	6.9	8.8251	0.22494	0.11247	–	10.95	76.4	83.658	2.49064	1.24532
	BSrMDy _{1.0} [11]	1420	–	–	–	480	–	178	–	25.276	–	572	–	3556	–	504.952	–
	PPbCZDy _{1.0} [57]	524	432	82.44	406.418	474	–	66.4	–	3.47936	2.86848	574	–	261.8	–	13.7183	11.3098
	PPbSrZDy _{1.0} [57]	507	397	78.3	546.505	474	–	128.7	–	6.52509	5.10939	567	–	242.6	–	12.2998	9.63122
	PPbMZDy _{1.0} [57]	509	413	81.13	456.671	474	–	65.1	–	3.31359	2.68863	574	–	253.4	–	12.8981	10.4654
	BSLZfDy _{1.0} [12]	202	165	80	1110.11	482	17.43	8.97	15.6347	0.18119	0.14801	575	14.86	37.3	55.4278	0.75346	0.61545
	TWTiDy _{1.0} [47]	700	630	90	158.73	484	16	19.1	30.56	1.337	1.2033	576	16	24	38.4	1.68	1.512
	B60LYDy _{1.0} [13]	1420	440	31	1568.5	483	–	1.9	–	0.2698	0.0836	575	–	13.3	–	1.8886	0.5852
	TPbWZDy _{1.0} [48]	183	160	87	785.519	483	18.39	11.4	20.9646	0.20862	0.1824	575	15.18	49.6	75.2928	0.90768	0.7936
	BPKAZfDy _{1.0} [14]	1797	542	30	1288.54	472	10.1074	2.27	2.29438	0.40792	0.12303	571	7.3883	28.9	21.3522	5.19333	1.56638
	BLSrZDy _{1.0} [15]	740	522	71	564.357	–	–	0.49	–	0.03626	0.02558	–	–	19.01	–	1.40674	0.99232
	BBiZDy _{1.0} [16]	463	264	57.01	1628.05	482	18.2	3.4	6.188	0.15742	0.08976	575	14.33	28.8	41.2704	1.33344	0.76032
	PPbZDy _{1.0} [58]	2671	–	–	–	475	21.31	12.5	26.6375	3.33875	–	586	16.21	8.81	14.281	2.35315	–
	BTAKDy _{1.0} [17]	1050	310	29.4	2273.43	6	4.1	2.46	0.4305	0.1271	–	–	5	56.6	28.3	5.943	1.7546
	BTCZDy _{1.0} [18]	647	298	44	1810.11	482	1.267	6.283	0.79606	0.40651	0.18723	575	14.176	4.992	7.07666	0.32298	0.14876
	BTBiCDy _{1.0} [18]	879	375	42	1529.01	482	2.304	16.285	3.75206	1.43145	0.61069	575	7.79	41.291	32.1657	3.62948	1.54841
	BTPbCDy _{1.0} [18]	883	311	35	2082.93	482	2.343	16.122	3.77738	1.42357	0.50139	575	8.38	38.599	32.346	3.40829	1.20043
	BTCCdDy _{1.0} [18]	865	279	32	2428.16	483	1.175	28.542	3.35369	2.46888	0.79632	575	7.987	43.996	35.1396	3.80565	1.22749
	BTcAMTiDy _{1.0} [19]	287	245	85	597.312	482	16.91	4.64	7.84624	0.13317	0.11368	575	15.91	36.59	58.2147	1.05013	0.89646
	BTBiZBfDy _{1.0} [20]	600	445.3	74.2	579.01	481	22	4.06	8.932	0.2436	0.18079	575	23.2	25.2	58.464	1.512	1.12216
	TBZofBfCdDy _{1.0} [49]	1114	372	33	1790.51	485	8.902	5.628	5.01005	0.62696	0.20936	577	7.225	46.341	33.4814	5.16239	1.72389
	BABfPfdy _{1.0} [21]	92.2	84	92	1058.77	483	19.5	2.29	4.4655	0.02111	0.01924	575	17.8	0.74	1.3172	0.00682	0.00622
	BPbAZDy _{1.0} [22]	212	132.8	62.6	2813.14	483	14.04	24.5	34.398	0.5194	0.32536	572	18.05	19	34.295	0.4028	0.25232
	BBiNCfDy _{1.0} [23]	1018	367	36	1742.48	482	14	3.146	4.4044	0.32026	0.11546	575	7.74	40.42	31.2851	4.11476	1.48341
	TPofZNDy _{1.0} [50]	374	335	90	311.278	482	14.81	6.09	9.01929	0.22777	0.20402	574	14.56	40.7	59.2592	1.52218	1.36345
	BPbSbDy _{1.0} [24]	1220	900	–	291.439	–	–	–	–	–	–	–	–	–	–	–	–
	SBNABaZDy _{1.0} [41]	1040	240	23	3205.13	479	–	2.3	0	0.2392	0.0552	572	–	18.8	–	1.9552	0.4512
	BTLoInDy _{1.0} [25]	1036	324	31	2121.17	–	–	–	–	–	–	–	–	–	–	–	–
	PNADy _{1.0} [59]	1107	1030	–	67.5314	–	–	–	–	–	–	–	–	–	–	–	–
	TWLFdY _{1.0} [51]	272	102	38	6127.45	483	15.47	4.01	6.20347	0.10907	0.0409	575	12.74	54.5	69.433	1.4824	0.5559
	TWNfDy _{1.0} [51]	201	97	48	5334.15	483	15.47	5.02	7.76594	0.1009	0.04869	575	12.74	77.2	98.3528	1.55172	0.74884
	TWKfDy _{1.0} [51]	190	137	72	2036.11	483	14.56	6.84	9.95904	0.12996	0.09371	575	12.74	81.4	103.704	1.5466	1.11518
	TWLFnDy _{1.0} [51]	181	157	87	844.565	483	14.56	6.66	9.69696	0.12055	0.10456	575	12.74	87.1	110.965	1.57651	1.36747
	TWKfNDy _{1.0} [51]	169	156	92	493.097	483	13.65	8.13	11.0975	0.1374	0.12683	575	12.74	94.4	120.266	1.59536	1.47264
	TWKfLDy _{1.0} [51]	161	153	95	324.768	483	14.11	7.97	11.2457	0.12832	0.12194	575	12.968	102	132.274	1.6422	1.5606
	BLCMBfNDy _{1.0} [26]	721	535	74	482.196	485.32	–	–	–	–	–	577.2	–	–	–	–	–
	BPbLBfNDy _{1.0} [26]	663	532	80	371.404	485.1	–	–	–	–	–	577.2	–	–	–	–	–
	BLCdBfNDy _{1.0} [26]	628	536	85	273.315	484.2	–	–	–	–	–	576.4	–	–	–	–	–
	BPbLCdBfNDy _{1.0} [26]	635	527	83	322.73	485.1	–	–	–	–	–	577.2	–	–	–	–	–
	BLMBfNDy _{1.0} [26]	472	441	93	148.93	485.8	–	–	–	–	–	577.2	–	–	–	–	–

(continued on next page)

Table 5 (continued)

Glass Label	$^4F_{9/2}$				$^6H_{15/2}$						$^6H_{13/2}$					
	τ_{rad}	τ_{exp}	η	W_{NR}	λ_p	$\Delta\lambda_{eff}$	$\sigma(\lambda_p)$	$\sigma(\lambda_p).\Delta\lambda_{eff}$	$\sigma(\lambda_p).\tau_{rad}$	$\sigma(\lambda_p).\tau_{exp}$	λ	$\Delta\lambda$	$\sigma(\lambda_p)$	$\sigma(\lambda_p).\Delta\lambda_{eff}$	$\sigma(\lambda_p).\tau_{rad}$	$\sigma(\lambda_p).\tau_{exp}$
BLCBfNdY _{1.0} [26]	477	442	93	166.007	485.8	–	–	–	–	–	577.2	–	–	–	–	–
SKNPfLdY _{1.0} [42]	536	370	69	837.031	486	13.01	6.3	8.1963	0.33768	0.2331	575	12.32	51.4	63.3248	2.75504	1.9018
SKNbZLfDy _{1.0} [43]	451	320	77	907.705	486	18	11.6	20.88	0.52316	0.3712	578	15.5	61.3	95.015	2.76463	1.9616
BPbTPBaZDy _{1.0} [27]	430	260	60	1520.57	486	9.12	9.71	8.85552	0.41753	0.25246	578	6.58	107.5	70.735	4.6225	2.795
PZNofAfDy _{1.0} [60]	666	396	59	1023.75	485.1	682.8	3	204.84	0.1998	0.1188	576.3	461.5	38	1753.7	2.5308	1.5048
BLGdBiDy _{1.0} [28]	475	423	89	258.803	–	–	–	–	–	–	–	–	–	–	–	–
BTCNbZDy _{1.0} [29]	999	162	–	5171.84	485	17.83	3.15	5.61645	0.31469	0.05103	577	17.06	32.25	55.0185	3.22178	0.52245
BTSrABcDy _{1.0} [30]	584	528	91	181.611	–	–	–	–	–	–	–	–	–	–	–	–
TSbWDy _{1.0} [52]	310	120	39	5107.53	483	16.07	4.45	7.15115	0.13795	0.0534	574	14.92	49.26	73.4959	1.52706	0.59112
BTNbfcDy _{1.0} [31]	155	142	91.61	590.641	485	17.65	8.16	14.4024	0.12648	0.11587	576	16.28	76.5	124.542	1.18575	1.0863
PGdNfDy _{1.0} [61]	437	540	123	-436.48	–	–	7.872	–	0.34401	0.42509	–	–	66.841	–	2.92095	3.60941
PGdNDy _{1.0} [61]	523	539	103	-56.758	–	–	4.654	–	0.2434	0.25085	–	–	66.841	–	3.49578	3.60273
PBiLNDy _{1.0} [62]	463	455	98.2	37.975	663	–	9	–	0.4167	0.4095	573	–	91.39	–	4.23136	4.15825
TBZDy _{1.0} [53]	619	384	62	988.658	–	–	–	–	–	–	–	10.25	67.65	69.3413	4.18754	2.59776
BPbBaZLfDy _{1.0} [32]	–	320	66	0	486	18	7.03	12.654	–	0.22496	577	16	54.35	86.96	–	1.7392
BSPbNADy _{1.0} [33]	765	623	81.38	297.947	482	16.8	32.27	54.2136	2.46866	2.01042	575	15.74	35.29	55.5465	2.69969	2.19857
BKNLfDy _{1.0} [34]	300	520	–	-1410.3	–	–	–	–	–	–	–	–	–	–	–	–
PLaCDy _{1.0} [63]	–	539	–	–	482	–	4.835	–	–	0.26061	574	–	57.814	–	–	3.11617
PLaCdY _{1.0} [63]	–	540	–	–	482	–	7.278	–	–	0.39301	574	–	62.466	–	–	3.37316
BSAZNfDy _{1.0} [35]	612.6	420.7	68.67	744.604	–	–	–	–	–	–	–	–	–	–	–	–
BMDy _{1.0} [36]	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
BBiBaDy _{1.0} [37]	–	322.51	–	–	481	20	4.8	9.6	–	0.1548	575	19	66.6	126.54	–	2.14792
BZABfCdY _{1.0} [38]	191	132	69	2340.16	486	10.9	26.5	28.885	0.50615	0.3498	576	12	158.3	189.96	3.02353	2.08956

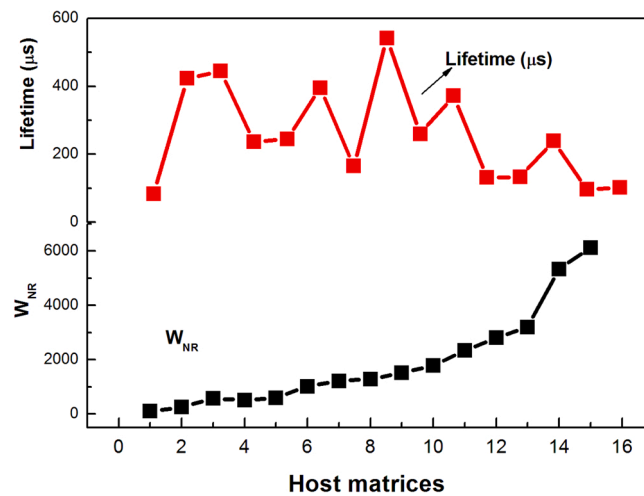


Fig. 7. Increasing order of non-radiative rates (W_{NR}) along with Dy^{3+} :glasses and corresponding experimental lifetimes (τ_{exp}). See text for glass compositions for X values from 1 to 15.

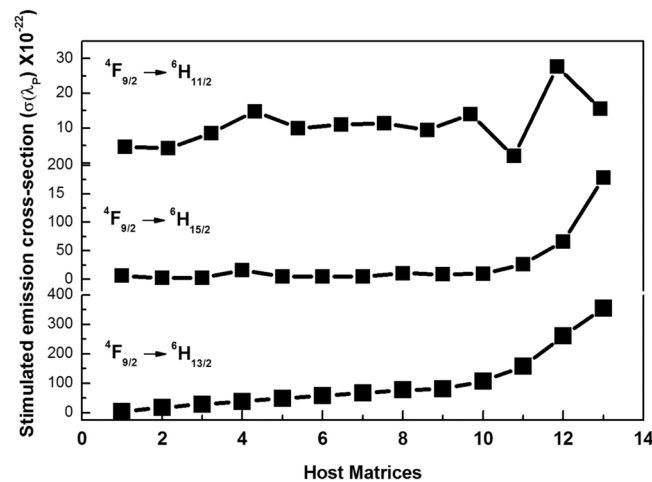


Fig. 8. Increasing order of stimulated emission cross-section for ${}^4F_{9/2} \rightarrow {}^6H_{13/2}$ transitions of 1.0 mol/% of Dy^{3+} :glasses. See text for glass compositions for X values from 1 to 13. Accordingly, stimulated emission cross-section for ${}^4F_{9/2} \rightarrow {}^6H_{15/2,11/2}$ transitions are arranged.

Fig. 8 shows the emission cross-section for ${}^4F_{9/2} \rightarrow {}^6H_{15/2,13/2,11/2}$ transitions of 1.0 mol% of Dy^{3+} :glasses. In x-axis, in the first instant, the compositions are arranged and fixed with increasing order of $\sigma(\lambda_p)$ for ${}^4F_{9/2} \rightarrow {}^6H_{13/2}$ transitions, then the values for ${}^4F_{9/2} \rightarrow {}^6H_{15/2}$ and ${}^4F_{9/2} \rightarrow {}^6H_{11/2}$ transitions are taken. The serial numbers 1–13 refers as 1. BTPbCDy_{1.0} [18], 2. SBNABaZDy_{1.0} [41], 3. BPAZfDy_{1.0} [14], 4. BABfPfDy_{1.0} [21], 5. TSbWDy_{1.0} [52], 6. PLACDy_{1.0} [63], 7. BTSrZofDy_{1.0} [PW], 8. PGdNfDy_{1.0} [61], 9. TN20ZofDy_{1.0} [46], 10. BPbTPBaZDy_{1.0} [27], 11. BZABfCdDy_{1.0} [38], 12. PPbCZDy_{1.0} [57] and 13. BSrMDy_{1.0} [11]. As seen from Fig. 8, it is interesting to note that $\sigma(\lambda_p)$ values follow the same trend for both ${}^4F_{9/2} \rightarrow {}^6H_{13/2}$ and ${}^4F_{9/2} \rightarrow {}^6H_{15/2}$ where as slightly scattered trend has been noticed for ${}^4F_{9/2} \rightarrow {}^6H_{11/2}$.

Fig. 9 shows the CIE chromaticity diagram for BTSrZofDy_{1.0} along with some of the Dy^{3+} :glasses such as TZNDy_{1.0} [46], BPKAZfDy_{1.0} [14], BTCZDy_{1.0} [18], TPofZNDy_{1.0} [50], PNADy_{1.0} [59] and BZABfCdDy_{1.0} [38]. As seen from Fig. 9, BTSrZofDy_{1.0} found to be in between lower [18,38,46] and higher [14,50,59] white light emission region. As can be seen, the 1.0 mol% Dy_2O_3 : BTSrZofDy_{1.0} found to be not only comparable to many systems but also possess superior luminescence properties to many of the systems and hence it could also be suitable for optical device applications. The luminescence characteristics of 66 data sets of Dy^{3+} -doped glasses presented in this paper are very much useful data bank for further work on rationalization of experimental and theoretical point of view to enhance dysprosium-ligand-radiation interaction simulation and systemization.

Table 6 shows the range of variation in optical and luminescence properties of 65 reported Dy^{3+} :glasses besides title glass which are collected in Table 5. As can be seen from Table 6, comparison of luminescence properties of ${}^4F_{9/2}$ level for 66 data sets of Dy^{3+} :glasses reveal that τ_{rad} (μs) are found to be shorter of 92 (BABfPfDy_{1.0} [21]) whereas longer of 2671 (PPbZDy_{1.0} [58]) and on the hand τ_{exp} (μs) are found to be shorter of 84 (BABfPfDy_{1.0} [21]) whereas longer of 1030 (PNADy_{1.0} [59]). Quantum efficiency (η , %) varied from 23

the uniform procedure and methodologies in preparation and characterization of different Dy^{3+} :glasses to arrive constructive conclusion for the optimization and design of optical devices.

4. Conclusions

Optical absorption and luminescence spectra and decay rates of Dy^{3+} -doped zinc strontium tellurofluoroborate glasses were analysed. Judd-Ofelt intensity parameters (Ω_λ , $\lambda = 2, 4, 6$) have been estimated and in turn radiative transition rates and lifetime, branching ratios for $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2,13/2,11/2}$ multiplet have also been predicted and compared with experimental values for title glasses. The relative intensity of emission peaks at 475 and 570 nm make the chromaticity coordinates lie at the region of white light. Also, the behavior of decay time profile changes from mono-exponential to bi-exponential (higher concentration) due to energy transfer through electric dipole-dipole interaction. Lifetime was evaluated for all the glass samples which are comparable to those obtained for oxide based glasses. CCT and CIE chromaticity values for titled glasses were located very near to white light region. Comparison of luminescence properties such as lifetimes, quantum efficiency, non-radiative rates of $^4\text{F}_{9/2}$ level as well as peak emission wavelengths, linewidths, emission cross-sections, gain bandwidth, optical gain and figure of merit for $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2,13/2}$ transitions of Dy^{3+} :glasses yields no systematic trend among these studied properties. It is summarized through the obtained values of branching ratio, emission cross-sections, CIE colour co-ordinates that the titled tellurofluoroborate glasses are also preferable for the design of visible laser gain media besides w-LEDs devices. Moreover, 66 data sets available in one place are highly useful to explore future rationalization attempts on luminescence properties of Dy^{3+} :glasses for the design WLED application.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

One of the authors, C.K. Jayasankar, is thankful to DAE-BRNS, Mumbai (No.2009/34/36/BRNS/3174 dated 12–02-2010) and UGC-BSR Faculty Fellow, New Delhi (No.F.18–1/2011 (BSR) dated 24–11-2017) for financial support. Also J. Kaewkhao would like to thank National Research Council of Thailand (NRCT) under research grants for talented mid-career researchers project (Project Number N41A640097).

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